

Forward operators: core operation and test error

Architecture	Core operation	RMSE (dB)	R^2	Correlation	Peak RMSE (dB)
DeepONet (DON)	Inner product	6.43	0.589	0.842	12.92
Deep Neural Operator (DNO)	Modulation (FiLM)	2.92	0.915	0.963	6.78
Deep Cat Operator (DCO)	Concatenation	2.70	0.927	0.967	6.70
Fourier Neural Operator (FNO)	Fourier transform	3.45	0.882	0.945	7.96
Wavelet Neural Operator (WNO)	Wavelet transform	3.46	0.881	0.943	9.05
Laplace Neural Operator (LNO)	Poles and residues	3.18	0.899	0.957	7.15
Graph Neural Operator (GNO)	Graph network	3.84	0.853	0.949	6.57
Set Transformer Operator (STO)	Attention	3.86	0.852	0.947	7.03
SIREN Neural Operator (SIREN)	Sine layers	3.12	0.903	0.958	6.96
Plain Neural Network (NN)	Dense layers	8.53	0.275	0.764	15.66

Latest 200-epoch models, 100k dataset (10,000 test spectra). Correlation: Pearson r over all points. Peak RMSE: error at the resonance peaks of the true ERP. Best value of each column in green.